FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
FULL TEST – XI
PAPER –2
TEST DATE: 28-05-2023

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B**.

Section – A *(01 – 04, 19 – 22, 37 – 40)*: This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

Section – A *(05 –10, 23 – 28, 41 – 46)*: This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – B *(11 – 18, 29 – 36, 47 – 54)*: This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.**

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	–1	In all other cases.

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Physics

PART – I

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

1. Li^7 target is bombarded with a proton beam of 10^{-4} A for 1hour to produce Be^7 of activity 1.8×10^8 disintegrations per second. Assuming that one Be^7 radioactive nucleus is produced by bombarding 1000 protons, determine its half life.(The isotope so formed is stable.)

(A) 8.63×10^6 s (B) 2.25×10^8 s

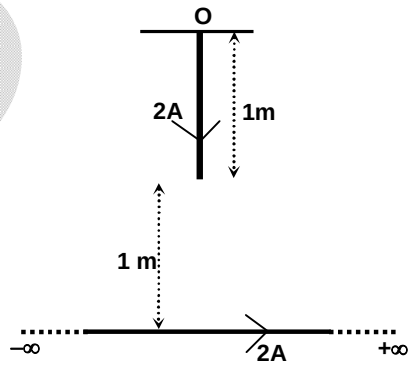
(C) 12.5×10^8 s (D) 6.45×10^6 s

2. A meteorite of mass m collides with a satellite perpendicular to the satellite velocity in its plane which was orbiting around a planet in a circular path of radius R. Due to collision, the meteorite sticks to the satellite (mass=10 m) and the satellite is seen to have gone into an orbit whose minimum distance from the planet is R/2. Determine the velocity u of the meteorite before collision, Mass of the planet is M.

(A) $\sqrt{\frac{49GM}{5R}}$ (B) $\sqrt{\frac{11GM}{242R}}$

(C) $\sqrt{\frac{58GM}{R}}$ (D) $\sqrt{\frac{11GM}{5R}}$

3. A wire of mass 0.1Kg and length 1m carrying a current of 2A is suspended from a point O. An another infinitely long wire carrying the same current is at a distance 1m below the lower end of the wire. The angular acceleration of the wire just after released from the position shown is…



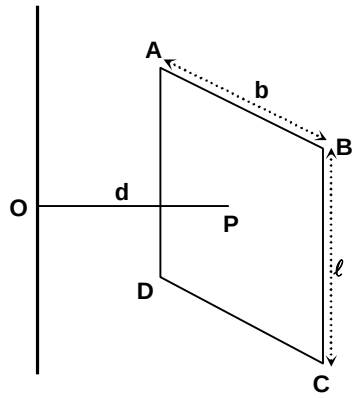
(A) $6.2\times10^{-8}\,rad/s^2$

(B) $2.1\times10^{-4}\,rad/s^2$

(C) $9.3\times10^{-6}\,rad/s^2$

(D) $4.5\times10^{-5}\,rad/s^2$

4. Find the electric flux crossing the wire frame ABCD of length ℓ width b and whose centre is at a distance OP=d from an infinite line of charge with linear charge density λ. Consider the plane of frame is perpendicular to line OP.



(A) $\frac{\lambda\ell}{\pi\varepsilon_o}\tan^{-1}\left(\frac{b}{2d}\right)$

(B) $\frac{\lambda\ell}{\pi\varepsilon_o}\tan^{-1}\left(\frac{\ell}{2b}\right)$

(C) $\frac{\lambda\ell}{2\pi\varepsilon_o}\tan^{-1}\left(\frac{d}{\ell b}\right)$

(D) $\frac{\lambda\ell}{2\pi\varepsilon_o}\tan^{-1}\left(\frac{b}{\ell d}\right)$

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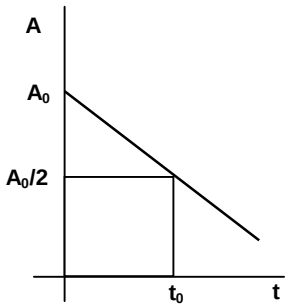
Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

5. A spherical shell is uniformly charged by a charge Q. A point charge q is placed at its centre. If the expansion of the shell is taking place from r to R, for this situation mark the correct statements.
   (A) If external force is acting, then work done by the external agent is negative.
   (B) If no external force is acting, then energy would be dissipated in this process.
   (C) If no external force is acting, then energy would be released in this process.
   (D) The work done by electrostatic force in this process is $\frac{Q(q+Q/2)}{4\pi\varepsilon_o}\left(\frac{1}{r}-\frac{1}{R}\right)$

6. A non conducting ring of mass m and radius R has a charge Q uniformly distributed over its circumference. The ring is placed on a rough horizontal surface such that plane of the ring is parallel to the surface. A vertical magnetic field $B = B_0t^2$ is switched on at t = 0. At t = 2s the ring just starts rotating about a vertical axis through its centre. Then the correct statement(s) is/are…
   (A) Friction coefficient between the ring and the surface is $\frac{2B_0RQ}{mg}$
   (B) Friction coefficient between the ring and the surface is $\frac{B_0RQ}{2mg}$
   (C) If the magnetic field is switched off after 4s, the angle rotated by the ring before coming to stop is $\frac{2B_0Q}{m}$
   (D) If the magnetic field is switched off after 4s, the angle rotated by the ring before coming to stop is $\frac{B_0Q}{m}$

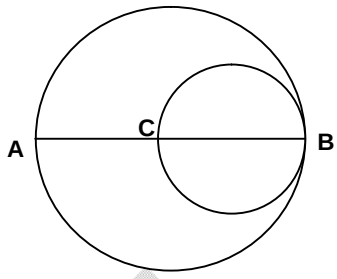
7. A radio active nuclide decays with a decay constant λ. Let $A=\log\left|\frac{dN}{dt}\right|$, where $\left|\frac{dN}{dt}\right|$ is the number of radioactive decays taking place per second. Suppose the experiment starts at t=0 and the curve between A and t is shown in figure, choose the correct statement(s) from the following.



   (A) Decay constant of radioactive sample is $\frac{A_0}{2t_0}$
   (B) Number of radioactive nuclei at the start of the experiment was $\frac{t_0}{A_0}e^{A_0}$
   (C) Number of radioactive nuclei after one half-life is $\frac{t_0}{A_0}e^{A_0}$
   (D) Half-life of the radioactive sample is t_0.

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8. A transparent sphere of radius R has a spherical cavity of radius R/2 as shown in the figure. μ is the refractive index of the material of sphere. A parallel beam of light is incident on the left surface, then…



(A) After the first refraction, image is formed at a distance $\frac{\mu R}{1-\mu}$ from C.

(B) After the first refraction, image is formed at a distance $\frac{\mu R}{\mu+1}$ from C.

(C) The possible refractive index of material of sphere, if the incident light has to finally focus at point B is $\frac{3+\sqrt{5}}{2}$

(D) The possible refractive index of material of sphere, if the incident light has to finally focus at point B is $\frac{3+\sqrt{3}}{2}$

9. Using Bohr model, for revolution of an electron in the nth orbit in an atom choose the correct dependency of following quantities on n and Z. (n is principle quantum number and z is atomic number)

(A) electric current $i_n \alpha \frac{Z^2}{n^3}$ (B) magnetic moment $M_n \alpha \frac{n}{Z^3}$

(C) Magnetic field at centre $B_n \alpha \frac{Z^3}{n^5}$ (D) angular speed $\omega_n \alpha \frac{Z^2}{n^3}$

10. Two heaters designed for the same voltage V have different power ratings. When connected individually across a source of voltage V they produce H amount of heat each in time t_1 and t_2 respectively. When they used together across the same source, they produce H amount of heat in time t, then..

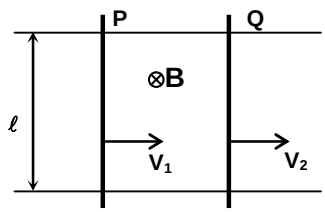
(A) If they are in series $t = t_1 + t_2$ (B) If they are in series $t = 2(t_1 + t_2)$

(C) If they are in parallel $t = \frac{t_1 t_2}{2(t_1 + t_2)}$ (D) If they are in parallel $t = \frac{t_1 t_2}{(t_1 + t_2)}$

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.***

11. Two infinitely smooth metal bars are placed parallel to each other in a horizontal plane. Two conducting rods P and Q of equal mass m and equal resistance R/2 are placed perpendicular to the bars. This arrangement is placed in a vertical uniform magnetic field B as shown in figure. At t = 0 P is given a velocity V_1 and Q is given a velocity $V_2 (< V_1)$. Assume the metal bars are perfectly conducting, the induced current passing through the rods is given by $i(t) = \frac{B\ell(V_1 - V_2)e^{-\left(\frac{YB^2\ell^2 t}{mR}\right)}}{XR}$. Find the value of X+Y.

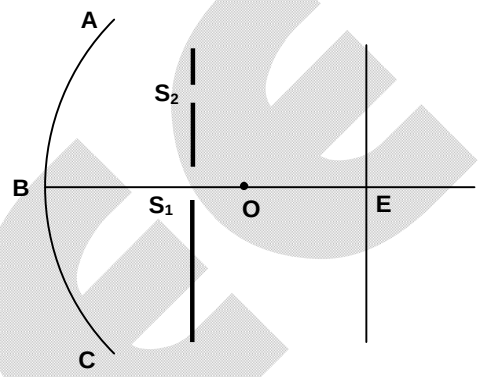


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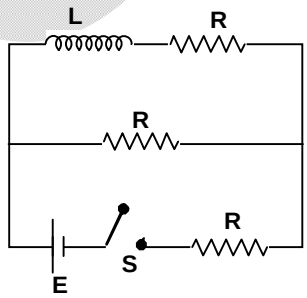
12. Two metal balls of same radius a are placed at a large distance apart in a medium of resistivity ρ. The resistance between the two balls is $\frac{\rho}{X\pi a}$. Find the value of X.

13. A lens of focal length f and aperture d forms an image of intensity I_0. If aperture of diameter d/2 in central region of lens is covered with a black paper, the focal length and intensity of image respectively becomes $\left(\frac{f}{x}\right)$ and $\left(\frac{yI_0}{z}\right)$. Find the value of x(y + z). (assume aperture to be << f)

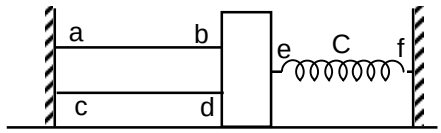
14. ABC is a spherical wavefront centered at O is incident on slits S_1 and S_2. $BS_1 = 3\lambda, S_1S_2 = 4\lambda,$ $BO = 6\lambda, S_1E = 128\lambda$, here λ is the wavelength of incident wave. A mica sheet of refractive index 1.5 is pasted on S_2. If the minimum value of thickness of mica sheet for which central maxima is formed at E is $\frac{31\lambda}{N}$.Find the value of N.



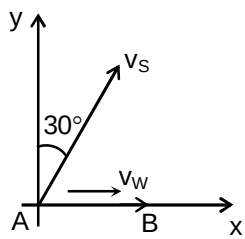
15. In the circuit shown, the switch is closed at t=0.The current drawn from the ideal cell as function of time is given by $\frac{E}{aR}+\frac{E}{bR}\left[1-e^{-\frac{cRt}{dL}}\right]$.Find the value of $\frac{ab}{cd}$.



16. A block of mass m placed on a smooth horizontal surface. It is connected by two elastic strings ab and cd of equal length ℓ, Young's Modulus Y & 2Y, area of cross section A & A/2 and a spring of spring constant K are all in unstretched condition initially. Given 2AY = 3KL. If the time period of small oscillations of the block along the horizontal surface in the plane of the diagram is $\frac{x\pi}{6}\sqrt{\frac{M}{K}}$, then x = ……

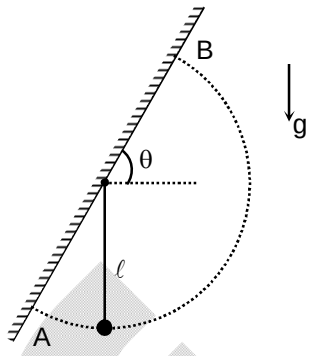


17. In the figure shown a source of sound of frequency 510 Hz moves with constant velocity $V_s = 20m/s$ in the direction shown. The wind is blowing at a constant velocity $v_w = 20m/s$ towards an observer who is at rest at point B. Find the percentage increase (to the nearest integer) in frequency detected by the observer corresponding to the sound emitted by the source at initial position A. [Speed of sound relative of air = $330m/s$]



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18. A small ball is hanging from a massless, inextensible thread of length ℓ. An inclined plane inclined at an angle 'θ' with the horizontal is passing through the point of suspension of the thread. The ball is projected horizontally with an initial speed 'v_0'.The ball hits the inclined plane elastically at points A and B and retraces the path at each collision. The maximum value of $\frac{KE_A}{KE_B} = \eta_1$ when $\theta = 30^o$. The maximum value of $\frac{KE_A}{KE_B} = \eta_2$ when $\theta = 60^o$ then $\left(\frac{\eta_1}{\eta_2}\right) =$



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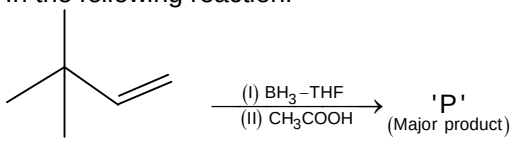
Chemistry

PART – II

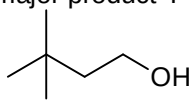
Section – A (Maximum Marks: 12)

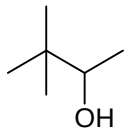
*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

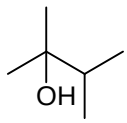
19. In the following reaction:

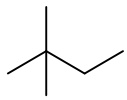


The major product 'P" is:

(A) 

(B) 

(C) 

(D) 

20. Sodium nitroprusside when reacts with sodium sulphide solution produces

(A) $Na_4\left[Fe(NH_3)_4\,NO\right]$

(B) $Na_2\left[Fe(CN)_5\,NO\right]$

(C) $Na_2\left[Fe(CN)_6\,NOS\right]$

(D) $Na_4\left[Fe(CN)_5\,NOS\right]$

21. Which of the following statement is incorrect?

(A) Neoprene is a polymer of isoprene
(B) Glyptal is a copolymer of ethylene glycol and phthalic acid
(C) Nylon-6, 6 is a condensation polymer
(D) Buna-S is copolymer of 1, 3-Butadiene and styrene

22. Which of the following statement is incorrect?

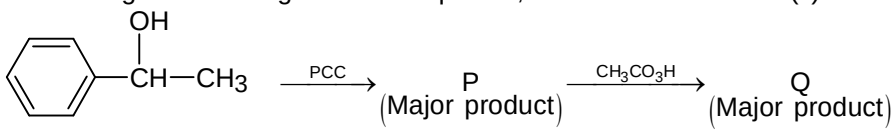
(A) Dipole moment of NH_3 is less than that of NF_3
(B) $N(CH_3)_3$ is more basic than $N(SiH_3)_3$
(C) BF_3 is a weaker Lewis acid than BCl_3
(D) Bond order of O_2^+ is greater than the bond order of O_2^-

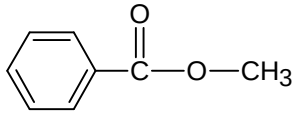
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Section – A (Maximum Marks: 24)

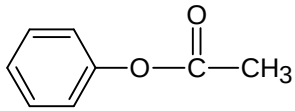
*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

23. Considering the following reaction sequence, the correct statement (s) is/are



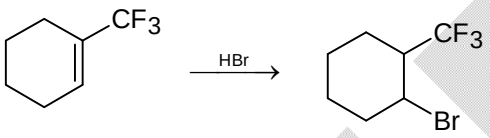
(A) Compound 'Q' is 

(B) Compound 'P' gives red precipitate with Fehling solution

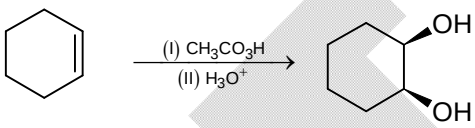
(C) Compound 'Q' is 

(D) Compound 'P' gives positive iodoform test

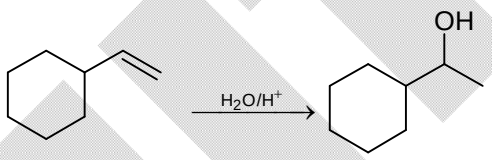
24. The reaction (s) in which the correct product(s) formed is/are

(A) 

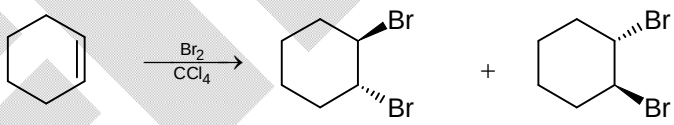
(Major product)

(B) 

(Major product)

(C) 

(Major product)

(D) 

25. The correct statement (s) regarding the crystalline structure of Zinc blende (ZnS) is/are

(A) Zn^{2+} ions are present in alternate tetrahedral voids

(B) S^{2-} ions are present in all octahedral voids

(C) S^{2-} ions are present at all the corners and all the face centre of the cube

(D) Effective number of Zn^{2+} ions per unit cell is 4

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26. Consider the following reaction given below. The reaction (s) in which the equilibrium position shift towards product side by adding inert gas at constant pressure is/are

(A) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

(B) $SO_2Cl_2(g) \rightleftharpoons SO_2(g) + Cl_2(g)$

(C) $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$

(D) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

27. The rate constant for a 1st order reaction is given by

$$\log k\left(\text{in } s^{-1}\right) = 2 - \frac{3\times10^3}{T}$$

Where k = Rate constant

T = Temperature

Given half life $\left(t_{1/2}\right) = 30$ min

R = 2 cal

The correct statement for the above reaction is/are

(A) Activation energy (Ea) = 13.818 kcal

(B) The value of rate constant (k) = 3.85×10^{-4} sec^{-1}

(C) Pre-exponential factor (A) = 10^2

(D) Pre-exponential factor (A) = 2

28. The correct statement(s) among the following statements is/are

(A) V_2O_5 is an acidic oxide

(B) N_2O is neutral oxide

(C) ClO_2 dimerises on cooling

(D) NO_2 is paramagnetic

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.***

29. 320 ml of $\frac{M}{10}KMnO_4$ solution in acidic medium when titrated required 112 ml of H_2O_2 solution. The volume strength of H_2O_2 solution is …….vol.

30. 1 mol of PCl_3 is dissolved in excess of water. Number of moles of NaOH which are required to neutralize this solution is

31. Equal masses of CH_4 and SO_2 were taken at 800 K and 400 K respectively. The ratio of their kinetic energies were found to be x : 1. The value of x is

32. 2.73 gram of $CoCl_3.6H_2O$ (molar mass 273 gm/mol) was dissolved in 100 gram water (freezing point 0^oC). The freezing point of solution was found to be -0.744^oC. K_f of water is 1.86 K kg mol^{-1}. If complete ionsation occurred, then the number of H_2O molecules in the coordination sphere of the compound $CoCl_3.6H_2O$ is

33. How many moles of $NaClO_3$ will be obtained when 9 moles of Cl_2 reacts completely with excess of hot and concentrated NaOH solution?

34. In a closed flask at 400 K NH_2COONH_4 (s) was taken. After some time a constant pressure of 3 bar was found in the flask. K_P for the reaction

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35. Benzaldehyde when treated with catalytic amount of CN^-, major product 'P" is formed. What is the number of π – bonds in one molecule of the product 'P'?

36. 0.5 gram of an organic compound having molecular formula $C_8H_{12}N_x$ was analysed by Kjeldahl's method. The ammonia evolved was treated with 50 ml of 0.2 M H_2SO_4 solution. The excess acid required 50 ml of 0.2 M NaOH for neutralization. The value of 'x' in the compound is

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Mathematics

PART – III

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

37. Let S_1 is the area bounded between the curve $y = \frac{1}{x}$, x-axis, the line $x = a_1$ and $x = b_1$. S_2 is the area bounded between $y = \frac{1}{x}$, x-axis, $x = a_2$ and $x = b_2$ where $a_1 < b_1 < a_2 < b_2$ and $x > 0$, if $a_1b_2 = b_1a_2$, then

(A) $S_1 - 2S_2 = 0$ (B) $2S_1 - S_2 = 0$

(C) $S_1 - S_2 = 0$ (D) $S_1 - 3S_2 = 0$

38. Let $f(x) = \frac{ax^2 + b}{\sqrt{x^2 + 1}}$ defined in $x \in R$ has minimum value equals to 3, then

(A) range of b is $[2, \infty)$ (B) range of b is $[3, \infty)$

(C) value of a for given b is $\frac{-b - \sqrt{b^2 + 9}}{2}$ (D) value of a for given b is $\frac{2b - \sqrt{b^2 - 9}}{2}$

39. Suppose there are 8 white balls and 2 yellow balls in a box. Each time one ball is drawn and replaced by a white one. Then the probability of drawing out all of the yellow balls just in the fourth draw is

(A) 0.1424 (B) 0.2423

(C) 0.3424 (D) 0.0434

40. The value of expression $\sqrt{\frac{1}{2}} \cdot \sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2}}} \cdot \sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2}}}}$ ..... is

(A) $\frac{2}{\pi}$ (B) $\frac{4}{\pi}$

(C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

41. Consider a bidirectional walk on x-axis. The particle starts (at time t = 0) from the origin and can make step of +1 with fixed probability p or –1 with probability q = 1 – p. If the probability that the particle is at x = r after n step is $P_n(r)$, then

(A) $P_9(3) = 84\ p^6q^3$ (B) $P_9(3) = 126p^6q^3$

(C) $P_{15}(3) = 5010p^9q^6$ (D) $P_{15}(3) = 5005p^9q^6$

42. Let define a sequence $T_1, T_2, T_3, ... T_n$ where $T_1 = 2$, $T_{n+1} = T_n^2 - T_n + 1$ for $n \geq 1$, then

(A) $T_n = 1 + \prod_{r=1}^{n-1} T_r$ (B) $T_n = 1 + \prod_{r=1}^{n-2} T_r$

(C) $\sum_{r=1}^{\infty} \frac{1}{T_r} = 2$ (D) $\sum_{r=1}^{\infty} \frac{1}{T_r} = 1$

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43. If ABC is a acute angle triangle, the

(A) $\sin\frac{A}{2}+\sin\frac{B}{2}+\sin\frac{C}{2}\le\frac{3}{2}$ (B) $\sin\frac{A}{2}+\sin\frac{B}{2}+\sin\frac{C}{2}\ge\frac{3}{2}$

(C) $\sin A+\sin B+\sin C\ge\frac{3\sqrt{3}}{2}$ (D) $\cos A+\cos B+\cos C\le\frac{3}{2}$

44. Let complex number $z = (a + \cos\theta) + (2a - \sin\theta)i$, where $|z| \le 2$, for any $\theta \in R$, then values of **a** are super set of

(A) $\left(-\infty, \sqrt{5}\right)$ (B) $\left[-\sqrt{5}, 0\right]$

(C) $\left[-\frac{1}{\sqrt{5}}, 0\right]$ (D) $\left[-\frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right]$

45. A function $f : (0, \infty) \to R$ satisfies,

(1) $f(a) = 1$ for a positive real number a

(2) $f(x)f(y)+f\left(\frac{a}{x}\right)\cdot f\left(\frac{a}{y}\right)=2f(xy)$ for any positive real number x and y, then

(A) f(2) + f(3) = 2 (B) f(2) + f(3) = 2a

(C) f(2) – f(3) = 0 (D) f(x) is strictly increasing in $(0, \infty)$

46. Line $x - 2y - 1 = 0$ and parabola $y^2 = 4x$ intersect at point A, B. Point C is on the parabola and $\angle ACB = 90°$, then coordinate of C is/are

(A) (1, –2) (B) $\left(2, 2\sqrt{2}\right)$

(C) (9, –6) (D) (9, 6)

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.***

47. The least value of the expression

$$f(x,t)=\left(x^2+\frac{1}{4x^2}\right)-2\left((1+\cos t)x+\frac{(1+\sin t)}{2x}\right)+(3+2\cos t+2\sin t)\ \forall\ x>0,\ t\in[0,2\pi] \text{ is}$$

48. The value of $\lim\limits_{x\to\infty} x^{-2}\int\limits_{x}^{x+1}\sin\left(t^2\right)dt$

49. The least value of the expression $(x + y)(y + z)$ where given that $x, y, z > 0$ and $xyz(x + y + z) = 1$

50. The number of positive integral solution of ordered pair (x, y, z, t) of the system $xz - 2yt = 3$, $xt + yz = 1$

51. Let $a_1, a_2, a_3 \ldots\ldots a_{100}, b_1, b_2, b_3 \ldots\ldots b_{100}$ be distinct real numbers, a matrix $[c_{ij}]_{100\times 100}$ is filled so that the number $a_i + b_j$ is element c_{ij}. If the product of all the element of each column is 1, then product of all the element of each row is k, then $\frac{1}{3^k}$ is equal to

52. Suppose the real number α and β satisfy the equation $\alpha^3 - 3\alpha^2 + 5\alpha = 1$ and $\beta^3 - 3\beta^2 + 5\beta = 5$, then the value of $\alpha + \beta$ is equal to

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53. Let the function $f(x) = a^{2x} + 3a^{x} - 2$ $(a > 0, a \neq 1)$ has maximum value 8 on interval $[-1, 1]$. If min value on this interval is k, then $4(k + 2)$ is equal to

54. Let P be a point on the curve $y = x + \frac{2}{x}$ $(x > 0)$ through P draw lines perpendicular to $y = x$ and y-axis with foot point A and B respectively. Then the value of $\left|\overrightarrow{PA} \cdot \overrightarrow{PB}\right|$ is

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